

S B VIJAY

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Education

Amrita Vishwa Vidyapeetham

Bachelor of Technology in Computer Science

Aug. 2023 – May 2027

Coimbatore, Tamil Nadu

Experience

Larsen & Toubro

Software Developer Intern

Apr. 2026 – May 2026

Chennai, Tamil Nadu, India

- Designed and built a **distributed remote software deployment system** enabling IT admins to silently install and uninstall packages across corporate Windows networks without physical machine access.
- Developed a deployment agent that is pushed to target machines, executes installations locally, maintains deployment logs, and self-removes from the endpoint upon job completion.
- Created a **Streamlit** dashboard to manage and track deployments across multiple machines, with support for network file storage, scheduled tasks, and secure credential-based authentication.
- Worked with enterprise Windows infrastructure including shared network storage, remote **PowerShell** execution, firewall configuration, and HTTP-based communication for reliable cross-network deployments.

Amrita School of Business | Pragati'26

Software Developer Lead

Jan. 2026 – Feb. 2026

Coimbatore, Tamil Nadu, India

- Led a **10-member** dev team to ship Pragati 2026 end-to-end; managed sprint planning, coordinated with organisers, and delivered registration, payment, and event management modules on schedule.
- Integrated **PayU** payment gateway with idempotent webhook validation and failure-handling logic, achieving zero duplicate transactions under peak concurrent registration load.
- Managed production infrastructure with reverse proxies and **Grafana + Prometheus** monitoring, maintaining uptime throughout live event operations.

Projects

ACM Open Source Platforms | Next.js, Go, PostgreSQL, Valkey, React, Tailwind CSS

[github](#)

- * Architected and shipped two open-source tracking platforms (Winter of Code Summer of Code) serving 1,000+ users across **100+ GitHub repositories**.
- * Engineered a real-time leaderboard using **Server-Sent Events**, delivering sub-second updates to 1,000+ concurrent users, eliminating polling overhead entirely.
- * Architected the frontend with **React** and **Zustand** for global state management with **GitHub OAuth**; configured **Valkey** caching on the backend and containerised the full dev environment with **Docker**.

CrisisOps AI | Google ADK, Gemini API, Flask, PostgreSQL, React, pgvector

[github](#)

- * Orchestrated **6 specialized agents** in a **Leader-Follower** pipeline using **Google ADK** to automate incident triage, Slack alerting, and post-mortem generation.
- * Implemented **pgvector** semantic search over **768-dim Gemini embeddings**; deployed on **Google Cloud Run** with Cloud SQL for elastic scalability.

Research & Publications

Benchmarking Deep Learning Models for Surgical Instrument Detection

Accepted – FACEIT 2026, July 2026

- * Constructed a novel dataset of **1,500 images** across **19 classes** via large-scale web scraping using BeautifulSoup.
- * Benchmarked YOLOv8/11/12, RF-DETR, and Roboflow v3; achieved peak detection accuracy of **90%**.

Technical Skills

Programming Languages: Java, Python, Go, TypeScript, C++ (familiar)

Frameworks & Libraries: Next.js, React.js, Node.js, Express.js, Gin, Flask, Elysia, React Native, Flutter, Tanstack, Zustand, Tailwind CSS

Databases & Storage: PostgreSQL, MySQL, Redis, Valkey, pgvector, Supabase, Snowflake, Drizzle ORM

Infrastructure & DevOps: Linux (bare metal), Google Cloud Run, Cloud SQL, AWS, Docker, Caddy, Grafana, Prometheus, ELK Stack

AI & ML: Google ADK, LangChain, MCP, YOLOv8/RF-DETR

Core CS: Data Structures & Algorithms, Distributed Systems Design, Cloud Computing